Swarm telepathy: a text-free channel between AI agents

Research summary · Playbox steer/ · 11 Nov 2026 · experiments E7 and E8

Summary

Two AI agents built on different model families can pass a concept to each other without sending any text. The sender reads private sentences about a topic and writes one vector. A fitted linear map translates it into the receiver's model space, and the receiver's model is steered with it while it continues neutral prompts. The receiving agent reads what its own model writes and names the topic, or answers "none".

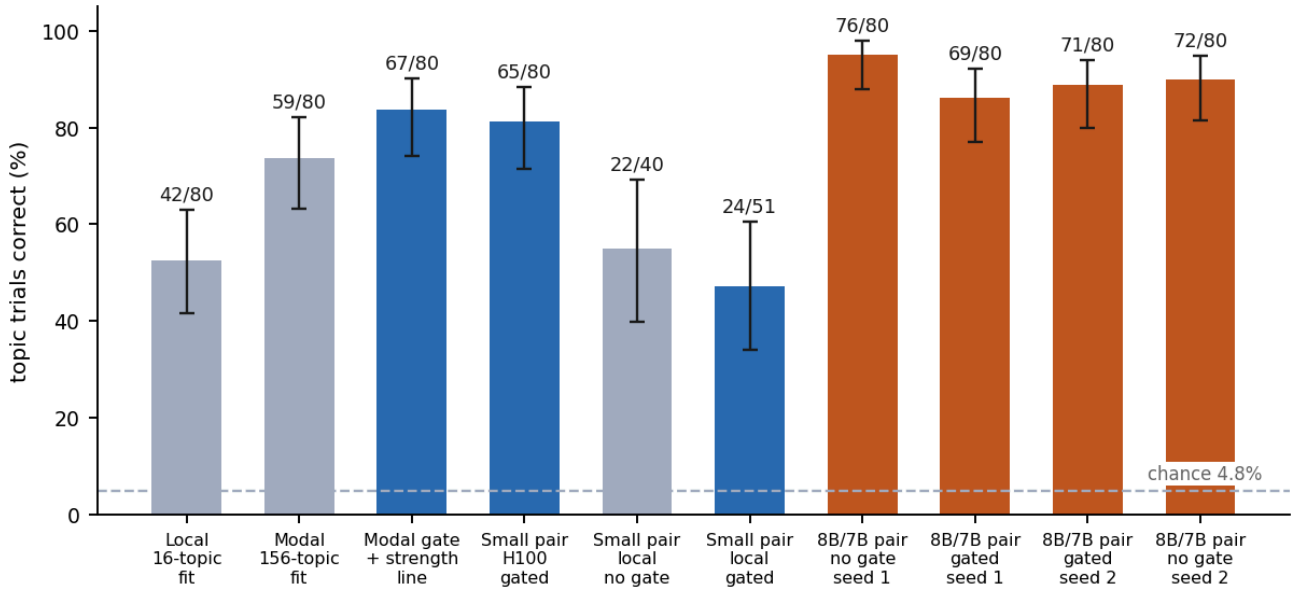
- **It works, far above chance.** On a blind test with 20 topics and a "none" option (chance 4.8%) a 135M / 1.7B pair named the topic in about 55% of trials locally and 81–84% on the H100 stack. A larger pair (Qwen3-8B-Base and OLMo-2-7B) on the same stack reached 140/160 = 87.5% (CI 81–92) gated (148/160 = 92% ungated), against 65/80 = 81.3% (CI 71–88) for the small pair with the same brain, gate recipe and controls: a gain of 6.2 points (one-sided exact p = 0.14), suggestive and not conclusive.
- **The hard part is the "nothing sent" case.** Receivers name a topic even when none was sent: 13 of 36 held-out controls on the small pair and 12 of 36 on the larger pair. Scale did not fix this.
- **A sender-side probe gate fixes it.** Small pair: false positives 2/25. Larger pair: 0/36 with topics at 69/80 = 86.3% (CI 77–92). The cost is some real topics muted to "none" and a weaker second direction. A second independent run of the gated larger pair gave 71/80 = 88.8% (CI 80–94) on topics and 1 false positive in 36 controls. Pooled over both gated runs: 140/160 = 87.5% (CI 81–92) on topics, 1 false positive in 72 controls (1.4%). The ungated repeat gave 72/80 = 90.0% (CI 81–95), so the gate's cost on topics is 5.0 points pooled over the two seeds of each arm.
- **It generalises across families.** A shared hub (E8) links five model families with one adapter pair per model and a 64-byte message; it scores within about 4 points of per-pair maps.
- **Scope.** Two model pairs, 20 coarse topics, one machine or node. A concept channel, not arbitrary-thought transfer.

1. Method

Sender (Agent A). Reads six concept sentences, takes the mean mid-layer hidden state (45% of depth), subtracts the mean over neutral sentences and maps the result with a ridge regression fitted on sentence pairs from both models (tokenizers differ, so pairs are whole sentences). Test topics are never in the fit set. **Receiver (Agent B).** The vector is added at 45% of depth while the local model continues neutral prompts. B sees only the continuations and a topic list and answers with one topic or "none". **Gate.** A ridge-logistic probe on the sender's unit difference vector, fitted on live Agent A text, scales the injected strength from 0 to 1; the strength travels with the vector and B is told to answer "none" when it is low.

Models. Small pair: SmolLM2-135M and Qwen3-1.7B-Base (held-out map cosine 0.89–0.95). Larger pair: Qwen3-8B-Base and OLMo-2-1124-7B, both width 4096, bf16 (held-out cosine 0.87 and 0.88). Agents are Hermes agents on Claude Haiku 5.5 (local) or Qwen3.8-27B (H100 node). **Protocol.** 20 topics × 2 repetitions × 2 directions = 80 topic trials, plus no-topic controls written live by Agent A (10 filler scenes and 8 off-list subjects per direction, none used in calibration). Agent files are kept outside B's directory, which holds only the vector file.

2. Results

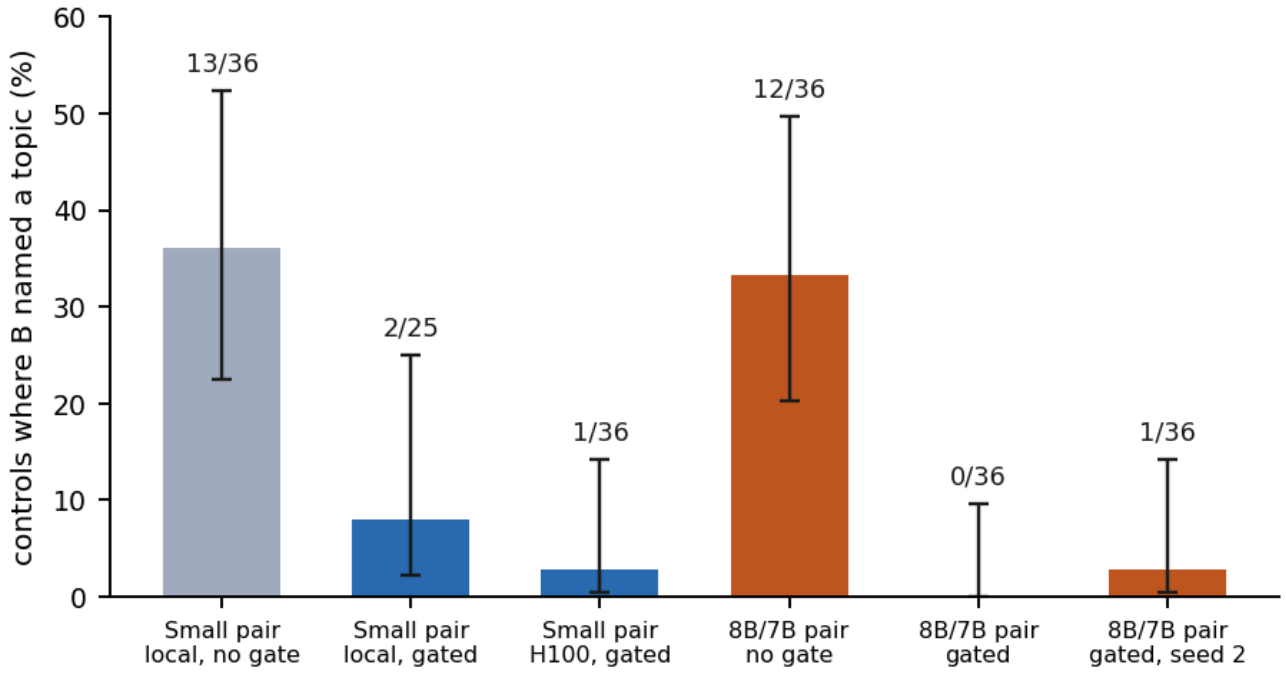


Topic accuracy, 95% Wilson intervals. Grey: no gate or earlier set-ups; blue: small pair; orange: larger pair (H100 stack, same brain as the blue "H100" bar). The first four bars changed fit data, strength and gating together and are not a controlled comparison.

Run (this week's protocol)	Topics correct	Controls where B named a topic
Small pair (135M / 1.7B), local, no gate	22/40 = 55.0% (CI 40–69)	13/36 = 36.1% (CI 22–52)
Small pair, local, probe gate (mixed plan)	24/51 = 47.1% (CI 34–60)	2/25 = 8.0% (CI 2–25)
Small pair, H100 node, probe gate (same brain and protocol)	65/80 = 81.3% (CI 71–88)	1/36 = 2.8% (CI 0–14)
8B / 7B pair, H100, no gate	76/80 = 95.0% (CI 88–98)	12/36 = 33.3% (CI 20–50)
8B / 7B pair, H100, probe gate	69/80 = 86.3% (CI 77–92)	0/36 = 0.0% (CI 0–10)
8B / 7B pair, probe gate, seed 2	71/80 = 88.8% (CI 80–94)	1/36 = 2.8% (CI 0–14)
8B / 7B pair, no gate, seed 2	72/80 = 90.0% (CI 81–95)	–

Controls are the same 36 held-out subjects per run (small gated: 25 controls from a shuffled mixed plan). Zero infrastructure failures in every run listed.

The "nothing sent" problem and the gate



Share of held-out controls on which B named a topic (lower is better).

- Early controls looked fine (8/8, 16/16) because they used one fixed neutral text, the same text the gate was calibrated on. On live filler and off-list subjects the receiver was wrong 9 of 20 and 6 of 12 times.
- A topic-ness gate looked good offline and failed live (topics fell to 46%); the calibration text had not come from the live agent.
- What held: a probe fitted on live Agent A text, with scenes and subjects held out whole in cross-validation, scored on fresh text. Off-list subjects still land on the nearest listed topic when they get through ("train delays" read as cars, "fishing" as ocean); that is a limit of a closed-list answer.

3. Larger model pair

Qwen3-8B-Base and OLMo-2-7B ran on one H100 next to a Qwen3.8-27B-FP8 brain, the same stack as the small-pair H100 run, so the pair is the main thing that changed. Ungated, topic accuracy was 148/160 = 92% over two runs; false positives did not shrink with scale (12/36, against 13/36 for the small pair). A probe calibrated on 248 live thoughts (held-out pass rates: topics 77/80 and 79/80, filler 0/24) brought false positives to 1 in 72 over two runs. Earlier comparisons of 55% (small pair, local, Haiku agents) with 95% (larger pair, H100, Qwen brain) mix two changes and should not be read as a scaling effect.

Run	Qwen3-8B → OLMo-2-7B	OLMo-2-7B → Qwen3-8B	Real topics muted to "none"
no gate	37/40 = 92%	39/40 = 98%	1
gate	30/40 = 75%	39/40 = 98%	4
gate, seed 2	32/40 = 80%	39/40 = 98%	4
no gate, seed 2	35/40 = 88%	37/40 = 92%	4

Pooled over the two seeds the gate costs the Qwen3-8B → OLMo-2-7B direction 72/80 = 90% → 62/80 = 78% (one-sided exact p = 0.026) and does not cost the other direction (76/80 = 95% → 78/80 = 98%, p = 0.90). The probe for the Qwen3-8B sender has the wider uncertain band (ramp 0.54 to 0.97), so it mutes more real topics than the OLMo-2 sender's (0.88 to 1.00).

Wrong-topic answers are mostly near neighbours (baseball / basketball / soccer, earthquake / ocean / volcano, cars / aviation).

4. Many families through one hub (E8)

Per-pair maps scale as N^2. E8 fits one shared hub (64 dimensions, 426 anchor sentences) with one adapter pair per model and sends a 64-byte int8 message per thought. It covers SmolLM2, Qwen3, Pythia, GPT-2-medium and TinyLlama. Mean over 20 ordered pairs, strength 0.5, chance 20%: hub 46.8%, per-pair 50.9%, receiver's own direction 80.5%, random vector 2.6%. Held-out cosine: hub 0.84–0.90, per-pair 0.88–0.95.

5. Reliability of the harness

Early large runs lost trials to infrastructure (32 of 86 timed out after a memory watchdog reboot; 51 of 86 failed on the first pass of the next run). The causes were a daemon serving one request at a time, orphaned agent processes after a timeout, and no retries. The harness now runs the daemon with queued requests and a health check, writes vectors atomically, runs each agent in its own process group and kills it on timeout, retries only infrastructure failures (never a valid wrong answer, so retries cannot inflate accuracy), and reports Wilson intervals with controls and directions separate. Every run in the results table had 0 infrastructure failures.

6. Limits

- Two model pairs and 20 topics; each configuration has one or two runs, and the larger-pair gain over the small pair on the same stack is not statistically established. Intervals are wide for the per-direction and per-topic splits.
- The small gated run is a 76-trial slice of a shuffled plan (51 topics, 25 controls), not the full plan.
- The probe is calibrated per sender and model; a new model needs a new calibration (about 250 agent runs, roughly 40 minutes on the H100).
- The vector is built from the sender's own sentences. This is a concept channel. Both ends sit on one machine or node; there is no hostile sender and the channel is not yet a Playbox room operation.
- Gate thresholds come from cross-validation on calibration text; controls are fresh subjects written by the same agent family, so a different agent could shift them.

7. Next steps

1. A third family and a larger receiver (for example a 27B) to test the scaling trend; each needs its own probe.
2. Targeted fit data for neighbouring topics, which cause most wrong answers.
3. Compare against published activation- and cache-communication methods (SwaT and LatentMAS are tracked in docs/steering.md, section 2d).
4. Wire the channel and its gate into a room as an audited operation.

Sources: docs/steering.md (E7 sections); steer/results/telepathy/*_private/results.json; steer/results/e8/e8.json. Generated by steer/tools/telepathy_report.py. Intervals are 95% Wilson.